torch.nn.utils.prune.random_structured#

https://pytorch.org/docs/stable/generated/torch.nn.utils.prune.random_struct

Description:

Prune tensor by removing random channels along the specified dimension. Prunes tensor corresponding to parameter called name in module by removing the specified amount of (currently unpruned) channels along the specified dim selected at random. Modifies module in place (and also return the modified module) by: adding a named buffer called name+'_mask' corresponding to the binary mask applied to the parameter name by the pruning method. replacing the parameter name by its pruned version, while the original (unpruned) parameter is stored in a new parameter named name+'_orig'. module (nn.Module) – module containing the tensor to prune

Function Signature

```
torch.nn.utils.prune.random_structured(module, name, amount, dim)[source]#

Parameters

Returns

Return type
```

Returns:

module (nn.Module)

Code Examples

```
>>> m = prune.random_structured(nn.Linear(5, 3), "weight", amount=3, dim=1)
>>> columns_pruned = int(sum(torch.sum(m.weight, dim=0) == 0))
>>> print(columns_pruned)
3
```

Detailed Documentation

Documentation

Prune tensor by removing random channels along the specified dimension.

Prunes tensor corresponding to parameter called name in module by removing the specified amount of (currently unpruned) channels along the specified dim selected at random. Modifies module in place (and also return the modified module) by:

adding a named buffer called name+'_mask' corresponding to the binary mask applied to the parameter name by the pruning method.

replacing the parameter name by its pruned version, while the original (unpruned) parameter is stored in a new parameter named name+'_orig'.

module (nn.Module) – module containing the tensor to prune

name (str) – parameter name within module on which pruning will act.

amount (int or float) – quantity of parameters to prune. If float, should be between 0.0 and 1.0 and represent the fraction of parameters to prune. If int, it represents the absolute number of parameters to prune.

dim (int) – index of the dim along which we define channels to prune.

modified (i.e. pruned) version of the input module